Lecture 6

The retarded kernel

Lecture 5 proves that a wave cannot propagate faster than unit speed. A source requires a further choice: its field must vanish before the source acts. We construct this solution for the wave operator in three spatial dimensions, with the convention $\Box = \partial_t^2 - \Delta$.

Objectives. *We prove the distributional identity for the retarded kernel, including its contribution at the cone vertex. We use the kernel to solve the wave equation for a smooth compactly supported source and determine where its field can occur.*

By the end of this lecture you will be able to:

- Verify the kernel identity by spherical averaging and integration by parts.
- Construct the unique past-vanishing C^2 solution for a smooth compact source.
- Bound the arrival times of the response to a source supported in a ball.

6.1 The support condition

We work on $\mathbb{R}^{1+3}$. A *test function* is a function $\psi \in C_c^\infty(\mathbb{R}^{1+3})$. A distribution is a linear functional on test functions, continuous when their supports stay in a fixed compact set and all their derivatives converge uniformly. Its pairing $\langle T, \psi\rangle$ is bilinear, without complex conjugation. Its support is the complement of the largest open set on whose tests it vanishes. The point distribution is given by $\langle \delta^4, \psi\rangle = \psi(0,0)$.

Definition 6.1 (Retarded fundamental solution)**.** A *fundamental solution* for $\Box$ is a distribution G such that

$$\langle G, \Box\psi\rangle = \psi(0,0) \qquad \text{for every } \psi \in C_c^\infty(\mathbb{R}^{1+3}).$$

It is *retarded* if its support lies in the future cone $J^+ = \{(t,x) : t \geq |x|\}$. It is *advanced* if its support lies in $J^- = \{(t,x) : t \leq -|x|\}$.

Distributional differentiation is defined by $\langle \partial_\mu T, \psi\rangle = -\langle T, \partial_\mu \psi\rangle$. Two derivatives remove this minus sign, so the defining identity says $\Box G = \delta^4$. The support condition selects the direction of time.

Theorem 6.2 (Retarded wave kernel)**.** *The distribution G_{ret} given by*

$$\langle G_{\mathrm{ret}}, \psi\rangle = \int_{\mathbb{R}^3} \frac{\psi(|x|,x)}{4\pi|x|}\,\mathrm{d}^3x, \qquad \psi \in C_c^\infty(\mathbb{R}^{1+3}), \tag{6.1}$$

is a retarded fundamental solution for $\Box$. Its support lies on $\{(t,x) : t = |x|\}$. For every $f \in C_c^\infty(\mathbb{R}^{1+3})$, the convolution $u = G_{\mathrm{ret}} * f$ is smooth and satisfies $\Box u = f$. It is the unique C^2 solution on all spacetime which vanishes for all sufficiently negative times.

The expression $\delta(t-|x|)/(4\pi|x|)$ is shorthand for (6.1). The displayed pairing gives its meaning at $x=0$. We prove the kernel identity first and the source assertions in Section 6.3.

6.2 Spherical averaging and the vertex

Write $\mathbb{S}^2=\{\theta\in\mathbb{R}^3:|\theta|=1\}$ and let $\mathrm{d}\Omega$ be its area measure, with total area 4π. Define the spherical mean by

$$M(t,r)=\frac{1}{4\pi}\int_{\mathbb{S}^2}\psi(t,r\theta)\,\mathrm{d}\Omega,\qquad t\in\mathbb{R},\quad r\geq 0.$$

Polar coordinates turn (6.1) into $\langle G_{\mathrm{ret}},\psi\rangle=\int_0^\infty rM(r,r)\,\mathrm{d}r$. The factor r makes the integral finite at the origin. If the spatial projection of $\operatorname{supp}\psi$ lies in $B_A(0)$, its absolute value is at most $A^2\sup|\psi|/2$. This bound proves continuity on tests with fixed compact support. Thus the formula defines a distribution. It vanishes on every test supported off $t=|x|$, which proves the asserted support condition.

To compute the wave operator, we must retain the angular part of the spatial Laplacian. The ball divergence formula (5.6) in Lecture 5, applied to the smooth vector field $\nabla\psi(t,\cdot)$, gives, for $r>0$,

$$4\pi r^2M_r(t,r)=\int_{B_r(0)}\Delta\psi(t,x)\,\mathrm{d}^3x.$$

Indeed, the outward normal derivative on ∂B_r is $\nabla\psi(t,r\theta)\cdot\theta$, whose integral is the left side. Differentiation in r by polar coordinates therefore gives

$$\frac{1}{4\pi}\int_{\mathbb{S}^2}\Delta\psi(t,r\theta)\,\mathrm{d}\Omega=\frac{1}{r^2}\partial_r(r^2M_r)=M_{rr}+\frac{2}{r}M_r.\tag{6.2}$$

This identity applies to every test function, whether or not it is radial. Time differentiation passes under the compact angular integral.

Set $H(t,r)=rM(t,r)$. The averaged wave operator then obeys

$$r\left(M_{tt}-M_{rr}-\frac{2}{r}M_r\right)=H_{tt}-H_{rr}.$$

Along the diagonal $t=r$, the chain rule cancels the mixed derivatives:

$$\frac{\mathrm{d}}{\mathrm{d}r}\big(H_t(r,r)-H_r(r,r)\big)=H_{tt}(r,r)-H_{rr}(r,r).$$

For $0<\varepsilon<A$, with A sufficiently large to contain the test's support, integration by parts yields

$$\int_\varepsilon^A r\left(M_{tt}-M_{rr}-\frac{2}{r}M_r\right)(r,r)\,\mathrm{d}r=\big[H_t(r,r)-H_r(r,r)\big]_\varepsilon^A.\tag{6.3}$$

The endpoint at A is zero, since the test and its derivatives vanish there.

At the other endpoint, smoothness of ψ makes M smooth for $r\geq 0$, with bounded derivatives near $(0,0)$. The integral of θ over the sphere is zero, so $M_r(t,0)=0$. Also $M(t,0)=\psi(t,0)$. Consequently

$$H_t(\varepsilon,\varepsilon)-H_r(\varepsilon,\varepsilon)=\varepsilon M_t(\varepsilon,\varepsilon)-M(\varepsilon,\varepsilon)-\varepsilon M_r(\varepsilon,\varepsilon)\longrightarrow-\psi(0,0).$$

The left integrand of (6.3) is the spherical mean of $r\Box\psi$ and is integrable at zero. Letting $\varepsilon\downarrow 0$ proves

$$\langle G_{\mathrm{ret}},\Box\psi\rangle=\psi(0,0).$$

The source is precisely the lower endpoint term. Solving the homogeneous equation away from the vertex would not establish this identity.

Example 6.3 (A normalisation check). Evaluate the pairing with the wave operator applied to $\psi(t,x) = e^{-(t-1/3)^2-|x|^2}$. This function is rapidly decreasing rather than compactly supported. The same integral and endpoint calculation apply because the function and its derivatives decay exponentially along the cone. Its spherical mean is $M(t,r) = e^{-(t-1/3)^2-r^2}$. Direct differentiation gives

$$\begin{aligned} r(\Box\psi)(r,r\theta) &= \left(\frac{40}{9}r - \frac{8}{3}r^2\right) e^{-(r-1/3)^2-r^2} \\ &= \frac{\mathrm{d}}{\mathrm{d}r}\left[\left(\frac{2}{3}r - 1\right) e^{-(r-1/3)^2-r^2}\right]. \end{aligned}$$

The upper endpoint is zero. The lower endpoint is $-e^{-1/9}$, so $\langle G_{\mathrm{ret}}, \Box\psi\rangle = e^{-1/9} = \psi(0,0)$. Jackson's kernel $\delta(t-r)/r$ would give $4\pi e^{-1/9}$. Our fundamental solution includes the divisor 4π.

6.3 The field produced by a source

For $f \in C_c^\infty(\mathbb{R}^{1+3})$, define its convolution with the kernel by

$$u(t,x) = \langle G_{\mathrm{ret}}(s,y), f(t-s,x-y)\rangle = \int_{\mathbb{R}^3} \frac{f(t-|y|,x-y)}{4\pi|y|}\,\mathrm{d}^3y. \tag{6.4}$$

For fixed (t,x), the function paired with G_{ret} is a test function of (s,y). This defines convolution without requiring a product of distributions.

We finish the proof of Theorem 6.2. Let (t,x) range over a compact set. Since the source has bounded spatial support, the integrand in (6.4) vanishes outside a fixed bounded set of y. Every derivative of f is bounded. Each differentiated integrand is therefore dominated by $C|y|^{-1}$ on a fixed ball. This is integrable in three dimensions. Differentiation under the integral proves $u \in C^\infty$. Second derivatives in (t,x) agree with second derivatives in (s,y) of $f(t-s,x-y)$. The kernel identity thus gives

$$\Box u(t,x) = \langle G_{\mathrm{ret}}, \Box_{s,y} f(t-s,x-y)\rangle = \langle \Box G_{\mathrm{ret}}, f(t-s,x-y)\rangle = f(t,x).$$

Choose $a \in \mathbb{R}$ with $f(s,z) = 0$ for $s < a$. For $t < a$ and every y, the retarded argument $t-|y|$ is below a. Hence $u(t,x) = 0$ for $t < a$.

For uniqueness, let v be another C^2 solution which vanishes for all sufficiently negative times. The difference $w = v - u$ solves $\Box w = 0$. Choose a slice $t = t_0$ before both fields can be nonzero. Both w and w_t vanish on that slice. The local cone uniqueness of Corollary 5.4, applied after translating time, gives $w = 0$ at each later point. For complex fields, apply that real-field result to the real and imaginary parts. Before this slice the fields already vanish. No finite total energy assumption is needed. This completes the theorem's proof.

Changing variables to the source position $z = x - y$ gives the spatial form

$$u(t,x) = \int_{\mathbb{R}^3} \frac{f(t-|x-z|,z)}{4\pi|x-z|}\,\mathrm{d}^3z. \tag{6.5}$$

The *retarded time* at the source position z, for observation at (t,x), is $t_{\mathrm{ret}} = t - |x-z|$. The travel time is the distance because the wave speed is one. The integral uses the source only at events on the observation point's past light cone, where $t - t_{\mathrm{ret}} = |x-z|$.

The cone in Figure 6.1 shows which source events can contribute.

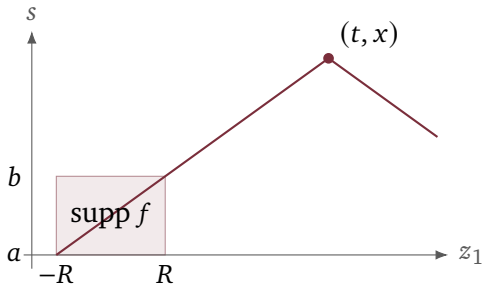


Figure 6.1: A spacetime section through one spatial axis of the source region and an observation point's past light cone. Only source events on the cone can enter the retarded integral.

Example 6.4 (A source in a ball)**.** Suppose a source is supported in

$$\operatorname{supp} f \subset [a,b] \times \overline{B_R(0)}, \qquad a < b, \quad R > 0.$$

For a fixed observer position x, the distance to a source position satisfies

$$\max\{|x| - R, 0\} \le |x - z| \le |x| + R.$$

A nonzero integrand in (6.5) requires $a \le t - |x - z| \le b$. Consequently the field can be nonzero only at times

$$a + \max\{|x| - R, 0\} \le t \le b + |x| + R. \tag{6.6}$$

For $a = 0$, $b = 2$, $R = 1$ and $|x| = 4$, the field vanishes when $t < 3$ and when $t > 7$. A wave from the near side needs at least three time units to arrive. A wave emitted at time two from the far side can arrive as late as time seven. These are support bounds, not claims that the field is nonzero at every intervening time.

The upper bound uses the largest distance to the source ball. Using its smallest distance would discard later arrivals from the far side. This finite response interval follows from the particular cone-supported kernel in three spatial dimensions.

6.4 Retardation is an additional condition

If $h \in C^2(\mathbb{R}^{1+3})$ solves $\Box h = 0$, then $u + h$ still solves $\Box(u + h) = f$. For example, $h = 1$ gives a field which does not vanish before the source. The equation alone therefore does not select the retarded solution. Local uniqueness shows that a nonzero homogeneous addition cannot vanish for all sufficiently negative times. The advanced kernel $G_{\text{adv}} = \delta(t + |x|)/(4\pi|x|)$ is also a fundamental solution, with support in the past cone.

The outgoing Helmholtz kernel of *Analysis*, Lecture 25, imposes a condition at spatial infinity for a fixed frequency. Retardation imposes a condition in time. Lectures 10 and 13 use the latter condition for fields produced by sources.

Summary. For $\Box = \partial_t^2 - \Delta$ on $\mathbb{R}^{1+3}$, the retarded kernel is defined by (6.1). Spherical averaging reduces its wave operator to an endpoint calculation. The vertex gives δ^4 with coefficient one. Convolution with a smooth compact source gives the unique C^2 solution vanishing sufficiently early. The source is evaluated at the retarded time $t - |x - z|$. Adding a homogeneous solution preserves the equation but can violate the past condition.

Self-check. Why can the term $2M_r/r$ not be dropped when averaging the Laplacian? Test the proposed omission on $\psi(t,x) = |x|^2$ near the origin.

Answer. The spherical mean is $M = r^2$. Its second radial derivative is 2, but $\Delta|x|^2 = 6$. The missing term $2M_r/r = 4$ supplies the difference. A compact cutoff equal to one near the origin makes this a test-function example.

Self-check. What is lost if the lower endpoint is omitted from (6.3)?

Answer. The remaining endpoint is zero. The calculation would incorrectly give $\Box G_{\mathrm{ret}} = 0$. The lower endpoint has limit $-\psi(0,0)$ and is subtracted, so it supplies $+\psi(0,0)$.

Self-check. A source is supported in $[1,3] \times \overline{B_2(0)}$. At a position with $|x| = 5$, give the earliest and latest possible response times.

Answer. The smallest distance is 3 and the largest is 7. Formula (6.6) permits a response only for $4 \le t \le 10$. Cancellations or smaller source support may narrow this interval.

Self-check. Does the uniqueness proof require the unknown C^2 solution to have finite total energy on each spatial slice?

Answer. No. Its difference from the retarded solution has zero data on an early slice. Lecture 5 applies the energy estimate on each bounded backward cone. The conclusion is pointwise uniqueness without a global integrability assumption.